Idea A: a p-uniform route for the relations case ($n = 2$, $T = \{1, 2\}$)

Notation as in (O1) of `mechanism_note.tex`. $F = \{1, \dots, \frac{p-1}{2}\}$, $\kappa_l(\delta) = 1 - \cos(2\pi l\delta/p) = 2s_l(\delta)^2$, $s_l(\delta) = |\sin(\pi l\delta/p)|$, $1 \le s \le \frac{3}{2}$, $r = 1/s$, $\tau = 3^{s-1}$, $\|a\| = (\sum_l a_l^r)^s$, $\Gamma = \Gamma_s(p)$, $m(a) = \min_{\delta\neq 0}(Ka)_\delta$, $\Delta(a)$ its minimisers, and the *efficiency* $e(a) = m(a)/(\Gamma\|a\|) \in [0, 1]$. For $0 \le v \le \tau a$ and phases θ,

$$Q(a, v, \theta) = \min_{\delta\neq 0}\Big[(Ka)_\delta + c(Kv)_\delta - \tfrac{1}{2}\operatorname{osc}_w D_\delta\Big], \qquad D_\delta(w) = \sum_l b_l \cos(2\pi lw/p + \psi_l), \;\; b_l = 2v_l s_l(\delta),$$

and $(Kv)_\delta = \sum_l b_l s_l(\delta)$. The relations case is closed iff $Q \le \Gamma\|a\|$ always. $c_* = 1/(2\sqrt{3})$ (Theorem A), $c_{12} = 1 - (2/3)^{s-1}$ (P_{12}).

1. Proved: reduction to a weighted Sidon constant

Definition. For $S \subseteq F$ and $\delta \neq 0$ let

$$\sigma_\delta(S) = \inf\Big\{\frac{\frac{1}{2}\operatorname{osc}_w \sum_{l\in S} b_l \cos(2\pi lw/p + \psi_l)}{\sum_{l\in S} b_l s_l(\delta)} \;:\; b \in \mathbb{R}^S_{\ge 0},\ \textstyle\sum b_l s_l(\delta) > 0,\ \psi \in \mathbb{R}^S\Big\}.$$

Theorem 1 (efficiency–Sidon reduction)**.** *Let $a \ge 0$, $S = \operatorname{supp} a$, $\delta_0 \in \Delta(a)$. If*

$$\sigma_{\delta_0}(S) \;\ge\; c - \frac{1/e(a) - 1}{\tau},$$

then $Q(a, v, \theta) \le \Gamma\|a\|$ for every $0 \le v \le \tau a$ and all phases.

Proof. Put $\sigma = \sigma_{\delta_0}(S)$; $\operatorname{supp} v \subseteq S$, so $\frac{1}{2}\operatorname{osc} D_{\delta_0} \ge \sigma(Kv)_{\delta_0}$ and $Q \le m(a) + (c - \sigma)(Kv)_{\delta_0}$. If $\sigma \ge c$ we are done since $m(a) \le \Gamma\|a\|$. Otherwise $(Kv)_{\delta_0} \le \tau(Ka)_{\delta_0} = \tau m(a)$ gives $Q \le m(a)(1 + (c - \sigma)\tau) \le m(a)/e(a) = \Gamma\|a\|$. □

Corollary 2. *(a)* (no arithmetic at all) *Every allocation with $e(a) \le 1/(1 + c\tau)$ is closed. $c\tau \le \frac{1}{2}$ for c_* and $c\tau = 3^{s-1} - 2^{s-1} \le 0.318$ for c_{12}, so the thresholds are $e \le \frac{2}{3}$ resp. $e \le 0.759$, for every p. (b)* (Parseval) *$\sigma_\delta(S) \ge (\sum_{l\in S} \kappa_l(\delta))^{-1/2}$, so a is closed if $\sum_S \kappa_l(\delta_0) \le (c - (1/e(a) - 1)/\tau)^{-2}$.*

Proof. (b) For mean-zero P, $\mathbb{E}P^2 \le (\max P)(-\min P) \le (\frac{1}{2}\operatorname{osc} P)^2$; $\mathbb{E}P^2 = \frac{1}{2}\sum b_l^2$ (distinct $l \in F$, $l \not\equiv -l'$); and $\sum b_l s_l \le (\frac{1}{2}\sum b_l^2)^{1/2}(2\sum s_l^2)^{1/2}$ with $2\sum s_l^2 = \sum \kappa_l$. □

So the whole problem is a lower bound for $\sigma_{\delta_0}(\operatorname{supp} a)$ *only for efficient allocations*, $e(a) > 1/(1 + c\tau)$, with the requirement relaxed by the coverage gap. At $e = 1$ (programme maximisers) the requirement is $\sigma \ge c$; Parseval gives it iff $\sum_S \kappa_l(\delta_0) \le 1/c^2$.

2. Proved: the tilted-mean bound (any phases)

Lemma 3 (tilted mean)**.** *Let $S \subseteq F$, $b \in \mathbb{R}^S_{\ge 0}$, $P(w) = \sum_l b_l \cos(2\pi lw/p + \psi_l)$, $\lambda > 0$, $\Pi(\lambda) = \prod_l I_0(\lambda b_l)$, $Z(\lambda) = \frac{1}{p}\sum_{j\in\mathbb{Z}_p} \exp\big(\lambda \sum_l b_l \cos(2\pi lj/p)\big)$. Then, for all phases ψ,*

$$\min(\max P, -\min P) \;\ge\; \frac{2\Pi'(\lambda) - Z'(\lambda)}{Z(\lambda)}, \qquad \min(\max P, -\min P) \;\ge\; \frac{\log\big(2\Pi(\lambda_2) - Z(\lambda_2)\big) - \log Z(\lambda_1)}{\lambda_2 - \lambda_1} \;\; (0 \le \lambda_1$$

Hence $\frac{1}{2}\operatorname{osc} P$ is at least either right-hand side, and $\sigma_\delta(S) \ge \inf_b \max\{(\frac{1}{2}\sum b_l^2)^{1/2}, \sup_\lambda \frac{2\Pi' - Z'}{Z}\}/\sum_l b_l s_l(\delta)$.

Proof. $M(\lambda) = \mathbb{E}_w e^{\lambda P(w)} = \sum_{k\in \mathrm{Rel}(S)} \prod_l I_{k_l}(\lambda b_l) e^{ik\cdot\psi}$, $\mathrm{Rel}(S) = \{k \in \mathbb{Z}^S : \sum k_l l \equiv 0\}$ (Jacobi–Anger and orthogonality). The $k = 0$ term is Π; with $\psi = 0$ the same sum of moduli is Z (the note's Laplace identity). Each $\prod_l I_{|k_l|}(\lambda b_l)$ is a power series in λ with non-negative coefficients, so it and its derivative are ≥ 0. Hence $M \leq Z$ and $M' \geq \Pi' - (Z' - \Pi')$. $M'/M = \mathbb{E}_{Q_\lambda} P \leq \max P$ for the probability $Q_\lambda \propto e^{\lambda P}$ on $\mathbb{Z}_p$, which gives the first bound ($\max P \geq 0$ anyway). $\log M$ is convex, so $(\log M(\lambda_2) - \log M(\lambda_1))/(\lambda_2 - \lambda_1) \leq (\log M)'(\lambda_2) \leq \max P$, and $M(\lambda_2) \geq 2\Pi(\lambda_2) - Z(\lambda_2)$, $M(\lambda_1) \leq Z(\lambda_1)$. For $-\min P$ replace ψ by $\psi + \pi$. □

With $x = \lambda b$: $Z/\Pi = 1 + \rho(x)$, where

$$\rho(x) = \sum_{k\in \mathrm{Rel}(S)\setminus 0} \prod_l \frac{I_{|k_l|}(x_l)}{I_0(x_l)} = \frac{\Pr_x(\mathbf{K} \in \mathrm{Rel}(S) \setminus 0)}{\Pr_x(\mathbf{K} = 0)}, \qquad \mathbf{K}_l \text{ independent Skellam}(x_l/2, x_l/2),$$

since $e^{-x} I_k(x)$ is the Skellam law. So $Z(x)e^{-\sum x}$ is the return probability to $0 \bmod p$ of the continuous-time walk with jumps $\pm l$ at rate $x_l/2$, and $\rho(x) = e^{E(x)}/p + \varepsilon(x)$ with $E(x) = \sum_l (x_l - \log I_0 x_l)$ and $\varepsilon(x) = e^{E(x)} \frac{1}{p} \sum_{j\neq 0} e^{-(Kx)_j} - 1$, the excess of the non-zero directions over *independent* phases ($\mathbb{E} e^{-\sum x_l \kappa} = e^{-E}$). Since $I_1/I_0 > \log I_0(x)/x$, the first bound is asymptotically $\sum_l b_l A(x_l)$, $A = I_1/I_0$, against $\sum_l \log I_0(x_l)/\lambda$ for the note's Proposition (tilt): at the programme scale it roughly doubles the certified fraction (0.386 vs 0.200 of $\sum b$ at $K = 1.5 \ln p$).

Proposition 4 (conditional, arithmetic-free certificate)**.** *Suppose* ($\mathrm{RM}_{\eta,L}$)*: $\rho(x) \leq \eta < 1$ for all $x \in \mathbb{R}^S_{\geq 0}$ with $E(x) \leq L$. Then for every b, ψ,*

$$\min(\max P, -\min P) \;\geq\; \Theta_{\eta,L}(b) := \sup_{\substack{0\leq \lambda_1 < \lambda_2 \\ E(\lambda_2 b) \leq L}} \frac{\sum_l \log \frac{I_0(\lambda_2 b_l)}{I_0(\lambda_1 b_l)} - \log \frac{1+\eta}{1-\eta} \mathbf{1}_{\lambda_1 > 0} + \log(1-\eta) \mathbf{1}_{\lambda_1 = 0}}{\lambda_2 - \lambda_1},$$

and $\sigma_\delta(S) \geq \theta_{\eta,L}(s(\delta)) := \inf_b \max\{(\frac{1}{2}\sum b_l^2)^{1/2}, \Theta_{\eta,L}(b)\}/\sum b_l s_l(\delta)$, a quantity depending only on the multiset $\{s_l(\delta)\}_{l\in S}$, η, L — not on p or on the arithmetic of S. Consequently (Theorem 1) a is closed if ($\mathrm{RM}_{\eta,L}$) *holds on* $\operatorname{supp} a$ *and $\theta_{\eta,L}(s(\delta_0)) \geq c - (1/e(a) - 1)/\tau$.*

Proof. E is coordinatewise increasing ($\frac{d}{dx}(x - \log I_0 x) = 1 - A > 0$), so $E(\lambda_1 b) \leq E(\lambda_2 b) \leq L$; use $Z \leq (1+\eta)\Pi$, $2\Pi - Z \geq (1-\eta)\Pi$ in Lemma 3, and $M(0) = 1$ when $\lambda_1 = 0$. □

The natural choice is $L = \ln p - \ln(1/\eta_0)$: then $e^E/p \leq \eta_0$ and (RM) only asks $\varepsilon \leq \eta - \eta_0$; with $\varepsilon \leq 0$ it says that a non-trivial Skellam vector is a relation of S with probability at most $1/p$, as for independent frequencies.

Proposition 5 (the $j = 0$ barrier)**.** *For all S, b, λ: $\frac{2\Pi' - Z'}{Z} \leq \min\{2, 2p\, e^{-E(\lambda b)}\} \sum_l b_l A(\lambda b_l)$, and the secant bound needs $E(\lambda_2 b) < \ln(2p)$. For b uniform on $K = \kappa \ln p$ frequencies every certificate of Lemma 3 is therefore $\leq 2A(x_\kappa)(1 + o(1)) \sum b$, $x_\kappa - \log I_0(x_\kappa) = 1/\kappa$; this is $\sim \sum b/\kappa$ for large κ.*

Proof. $Z \geq \Pi$, $Z \geq e^{\lambda \sum b}/p$ (the $j = 0$ term), $Z' \geq 0$, $\Pi' = \Pi \sum bA$. For uniform b the factor $\min\{2, 2pe^{-Kg(x)}\}$, $g(x) = x - \log I_0 x$, kills every x with $Kg(x) \geq \ln p + t$, $t \to \infty$, $t = o(\ln p)$; the secant bound is $\leq (\log \Pi)'(\lambda_2) = \sum bA(\lambda_2 b)$ by convexity of $\log \Pi$. □

The same $j = 0$ argument caps positive-definite product tests $G = \prod_l g_l(2\pi l w/p + \psi_l)$ ($g_l \geq 0$, $\hat{g}_l \geq 0$, $\hat{g}_l(0) = 1$): $\max P \geq (\text{main} - |\text{relation terms}|)/Z_G$ with $Z_G = \frac{1}{p}\sum_j \prod_l g_l(2\pi l j/p) \geq \max\{1, \prod_l g_l(0)/p\}$ and main $\leq \sum_l b_l \hat{g}_l(1) \leq \sum_l b_l \frac{g_l(0)-1}{2}$; for uniform factors this is $\leq \frac{e^{1/\kappa}-1}{2} \sum b \sim$

$\sum b/(2\kappa)$. So every argument that bounds relation terms by their moduli is capped at order $(\ln p/K)\sum b$, while the large-deviation value for generic phases is of order $(\ln p/K)^{1/2}\sum b$. Beating the cap needs cancellation among relation terms, i.e. phase information.

Lemma 6 (thin frequencies pay for themselves)**.** *Let $a = a' + a''$ with disjoint supports and $a''_l \le \vartheta\|a'\|$ for all l, $\vartheta = \big(\Gamma s/(2(1+\tau+c\tau))\big)^{s/(s-1)}$. If a' is closed (for all $v' \le \tau a'$, all phases), then a is closed.*

Proof. $D_\delta = D'_\delta + D''_\delta$ with $|D''_\delta| \le \sum_l b''_l$, so at the direction δ that closes (a', v', θ), $Q(a, v, \theta) \le \Gamma\|a'\| + \sum_l (a''_l\kappa_l + cv''_l\kappa_l + 2v''_l s_l) \le \Gamma\|a'\| + 2(1+c\tau+\tau)\sum_l a''_l$. By convexity of $t \mapsto t^s$, $\Gamma\|a\| \ge \Gamma\|a'\| + \Gamma s\|a'\|^{1-r}\sum_l (a''_l)^r$, and $2(1+c\tau+\tau)a''_l \le \Gamma s\|a'\|^{1-r}(a''_l)^r$ iff $a''_l \le \vartheta\|a'\|$. □

At $s = \frac{3}{2}$ with $\Gamma \approx \gamma^* K^{-1/2}$ and typical core weight $\|a'\|K^{-3/2}$, the threshold is $(0.23\gamma^*)^3 \approx 10^{-3}$ core weights, independent of p; but the number of frequencies above it can still be $\sim 10^2 K^*$, so Theorem 1 (which ignores the box $v \le \tau a$) must be applied to a core, not to $\operatorname{supp} a$.

Proposition 7 (no universal Laplace bound)**.** *The identities $\sum_{j\in\mathbb{Z}_p} \kappa_l(j) = p$ and $\sum_j \kappa_l(j)\kappa_k(j) = p(1 + \frac{1}{2}[l = k])$ $(l, k \in F)$ hold for every support. For $S = \{1, \dots, K\}$ and $x_l \equiv x \in (0, 1]$, $\frac{1}{p}\sum_{j\neq 0} e^{-(Kx)_j} \ge \big(\frac{2}{\pi K^{3/2}} - \frac{2}{p}\big)e^{-2x}$, while $e^{-E(x)} \le e^{-3Kx/4}$. Hence no bound for the averaged exponential coverage that uses only properties every allocation has can give* (RM)*; the hypothesis must use near-optimality ($e(a) > 1/(1+c\tau)$).*

Proof. $(Kx)_j \le x\sum_{l\le K} 2\pi^2 l^2 j^2/p^2 \le 2\pi^2 x K^3 j^2/p^2 \le 2x$ for $|j| \le p/(\pi K^{3/2})$. $e^{-x}I_0(x) \le e^{-x+x^2/4} \le e^{-3x/4}$ on $[0, 1]$. □

What survives of "coverage ⇒ Sidon". Min-coverage C bounds every $j \neq 0$ term by e^{-C} (the note's form), which needs $C \ge E$, i.e. coverage fraction $\gtrsim 0.77$–0.8 at the relevant scale; programme maximisers have 0.59–0.79 ($p \le 1009$) and the LD model predicts $\to 0.46$. The averaged form is exactly the Skellam return probability, and that is what has to be controlled. Transference forces short relations, but their Skellam weight $\prod_l I_{|k_l|}/I_0 \le A(x)^{\|k\|_1}$ is small; what matters is their number, i.e. ε.

3. Numerical (files in this directory)

All at $s = \frac{3}{2}$ ($b = 2$). Supports: `supports.json` ($p \le 1009$: best of `relations2/supports_s1.5.json`, the sibling `ideaC` runs, exhaustive search for $p \le 29$); $p \ge 2003$: local search plus reweighted LP (`largep.py`, `refine.py`), giving near-optimal allocations for several support sizes K per prime. Efficiencies use the best Γ found, which can only overstate e and the L^2 ratio; so passes are conservative, failures are not.

N1. Programme maximisers, $5 \le p \le 1009$ (`cert.py`, `cert_prog.out`, `relratio.out`).

p	29	61	101	151	211	307	401	503	601	701	809	1009
K^*	4	5	7	6	8	9	8	9	12	10	11	11
$\sum_S \kappa(\delta_0)$	3.03	3.26	4.28	3.37	4.62	5.29	4.50	4.98	7.43	5.53	5.91	5.84
σ_P	0.57	0.55	0.48	0.55	0.47	0.43	0.47	0.45	0.37	0.42	0.41	0.41
tilt/$\sum bs$	0.42	0.56	0.50	0.60	0.44	0.41	0.53	0.49	0.40	0.44	0.44	0.49
$\max R$	0.48	0.50	0.79	0.51	0.78	0.81	0.72	0.84	0.94	0.86	0.94	0.83

At every one of the 33 primes, $\sigma_{\delta_0} \geq \sigma_P \geq 0.367 > c_*$, so Theorem 1 closes the vertex under Theorem A. The tilted mean at the vertex direction ($b \propto ys$) is comparable to Parseval (0.30–0.60 for $p \geq 23$); its infimum over b is attained near the Parseval-extremal direction $b \propto s$, so at these sizes it adds ≤ 0.01. Relation mass: $R = \rho(x)/(e^{E(x)}/p)$ at the binding directions and scales $x_{\max} \in \{\frac{1}{2}, 1, 2\}$ is ≤ 0.97 for every support ($23 \leq p \leq 10007$), median 0.07–0.61; random K-sets with the same profile have median 0.12–0.87 and 90% quantile up to 1.38. So $\varepsilon < 0$: programme supports carry fewer weighted relations than independent frequencies.

N2. Spread near-optimal allocations (`certlarge.py`, `cert_large_*.out`). Columns: efficiency e, water-filling ratio C_{L2} (Theorem A / P_{12}; ≤ 1 closes), Parseval index and σ_P, requirement $c - (1/e - 1)/\tau$ of Theorem 1, and $\Gamma_{\mathrm{crit}} = \min_\delta[(Ka)_\delta + \tau g_\delta(a)]/\|a\|$ (the L^2 route closes a iff $\Gamma_s(p) \geq \Gamma_{\mathrm{crit}}$). Best Γ found: 0.16425, 0.14784, 0.13999, 0.12544 for $p = 2003, 5003, 10007, 30011$ (the last two after swap refinement, which raised them by 2.4% and 2.1%; the heuristic still undershoots, so passes are safe and failures are within the uncertainty). The tilt adds little at these sizes: box-aware tilt (`boxtilt.py`) vs water-filling at $p = 30011$: 0.973 vs 0.988 ($K = 23$), 1.006 vs 1.021 ($K = 26$), 1.032 vs 1.032 ($K = 30$). Under P_{12} the tilt alone closes $p = 30011$, $K = 30$. (The CMA estimates of $\inf_b$ of the tilt certificate in `cert_large_*.out`, column sigT, are upper estimates and are contradicted by the box searches; they are not used.)

p	K	e	C_{L2}(A)	$\mathrm{C}_{L2}(\mathrm{P}_{12})$	$\sum_S \kappa(\delta_0)$	σ_P	need(A)	need(P_{12})	Γ_{crit}(A)
2003	12	1.000	1.000	1.000	6.5	0.391	0.289	0.184	0.1643
2003	14	0.981	0.981	0.981	7.9	0.355	0.277	0.172	0.1611
2003	16	0.969	0.969	0.969	9.7	0.321	0.270	0.165	0.1591
2003	18	0.956	0.956	0.956	11.7	0.293	0.262	0.157	0.1570
2003	20	0.946	0.960	0.946	13.1	0.276	0.256	0.150	0.1576
5003	15	0.984	0.984	0.984	8.1	0.351	0.279	0.174	0.1455
5003	17	0.990	0.990	0.990	9.9	0.317	0.283	0.178	0.1463
5003	19	0.981	0.981	0.981	10.6	0.307	0.278	0.173	0.1451
5003	22	0.943	0.968	0.943	13.9	0.268	0.254	0.149	0.1430
10007	18	0.972	0.972	0.972	10.0	0.316	0.272	0.167	0.1360
10007	20	0.963	0.963	0.963	11.6	0.294	0.267	0.161	0.1348
10007	23	0.958	0.992	0.958	14.2	0.265	0.263	0.158	0.1389
10007	26	0.933	0.996	0.933	16.7	0.245	0.247	0.142	0.1395
30011	23	0.973	0.988	0.973	13.0	0.278	0.273	0.168	0.1240
30011	26	0.975	1.021	0.975	15.6	0.254	0.274	0.169	0.1281
30011	30	0.948	1.032	0.948	18.7	0.231	0.257	0.152	0.1294

N3. True room (`truesig.py`; adversarial search over (b, ψ), calibrated: it finds 0.18 for the progression $\{1, \ldots, 26\}$, close to the Parseval floor 0.14). For the spread allocation $p = 5003$, $K = 22$: $\sigma \leq 0.69$; for $p = 10007$, $K = 26$: $\sigma \leq 0.66$; needed 0.25–0.26, Parseval gives 0.27, 0.245.

N4. Large-deviation model (`asym*.py`): $p \to \infty$, $K = \kappa \ln p$, binding-direction law of $\kappa_l(\delta_0)$ = arcsine law tilted to the LD minimum over p directions, uniform weights, relation excess ≤ 0. The programme optimum of the model is at $\kappa^* \approx 3.2$, $\gamma^* \approx 0.46$.

$\kappa = K/\ln p$	2	3.2	4	5	6	7	8	9	10	13	17	20
coverage γ	.340	.464	.516	.564	.600	.629	.652	.671	.688	.725	.759	.778
efficiency e	.928	1	.996	.973	.946	.917	.889	.863	.839	.776	.710	.671
need, Thm A	.244	.289	.286	.273	.255	.236	.217	.197	.178	.122	.053	.006
need, P_{12}	.138	.184	.181	.168	.150	.131	.112	.092	.073	.017	< 0	< 0
tilt, $\inf_b$	.749	.388	.288	.217	.175	.145	.123	.107	.095	.070	.052	.044
LD truth ($b \propto s$)	1.45	1.06	.916	.797	.713	.650	.600	.560	.526	.452	.388	.355

(All entries are fractions of the payment basis $\sum_l b_l s_l(\delta_0)$.) Under P_{12} the tilt certificate beats the requirement at every κ (minimum ratio 1.10 near $\kappa = 8$); under Theorem A it fails for $4 \lesssim \kappa \lesssim 17$. The heuristic true value (generic phases) has a margin $\geq 3\times$ everywhere.

4. The exact remaining gap

(G1) *The arithmetic input.* Prove ($\mathrm{RM}_{\eta,\ln p - L_0}$) for the (cores of) efficient allocations, $e(a) > 1/(1+c\tau)$: a non-trivial Skellam vector on $\operatorname{supp} a$ is a relation mod p with probability $\leq (1+\eta')/p$ times the trivial one. Proposition 7 shows that no property shared by all allocations implies it (arithmetic progressions violate it by a factor $p^{\Omega(1)}$); it must come from near-optimality. Numerically $R = \rho/(e^E/p) \leq 0.97$ on every programme support ($23 \leq p \leq 10007$), below random sets of the same size.

(G2) *Theorem A is not enough for any modulus-based argument.* In the large-deviation model the tilt certificate falls below $c_* - (1/e - 1)/\tau$ for $4 \lesssim K/\ln p \lesssim 17$ (spread allocations of efficiency 0.71–0.99), and Proposition 5 says no tilt or positive-definite product test can do better than order $\ln p/K$. Closing this range with Theorem A needs a bound that uses cancellation among relation terms (phases), i.e. a flat-polynomial theorem for $O(\log p)$-element sets: $\sigma_{\delta_0}(S) \geq c_* - (1/e-1)/\tau$. Numerically the true σ has room (adversarial search: $\sigma \approx 0.66$–0.69 against 0.25–0.26 needed), so the inequality is not tight; only the proofs are.

(G3) *Under P_{12}* the model certificate exceeds the requirement for every $K/\ln p$ (minimum ratio 1.10 at $K/\ln p \approx 8$, efficiency 0.89). So P_{12} + (G1) + the model's description of efficient allocations (binding-direction law, uniform weights) gives a p-uniform closure; turning the model into a theorem is the remaining work, and the 10% margin is thin.

(G4) *Halo.* Theorem 1 takes the infimum over all amplitude directions on $\operatorname{supp} a$ and ignores $v \leq \tau a$. Lemma 6 removes frequencies below $\approx 10^{-3}$ core weights, p-uniformly; between 10^{-3} and 1 core weight a box-aware version (water-filling with the tilt in place of Parseval) is needed. It is computable (`boxtilt.py`) but not yet analysed in the model.

(G5) *Finite p, Theorem A.* The L^2 (water-filling) criterion holds on every allocation tested up to $p = 10007$ (≤ 0.996 with the best Γ found) and fails at $p = 30011$ for spread near-optimal allocations ($K = 26, 30$, $e = 0.95$–0.98, ratio 1.021, 1.032) unless $\Gamma_s(30011) \geq 0.1294$, 3.2% above the best value found (0.12544, which was still rising by $\sim 1\%$ per refinement round). The tilt with exact relation mass helps only marginally (box-aware certificate 0.973 at $K = 23$, 1.006 at $K = 26$, 1.032 at $K = 30$). So under Theorem A the relations case is numerically closed up to $p \approx 10^4$ and the L^2 route is at its edge around $p \approx 3 \cdot 10^4$; beyond, (G2) is the obstruction.

(G6) *Finite p, P_{12}.* $\mathrm{C}_{L2} \leq 0.99$ on every allocation tested up to $p = 30011$, and at $p = 30011$ the tilt *alone* (no Parseval) already closes the most spread allocation numerically ($K = 30$; box

search finds no residual gain at three binding directions). So under P_{12} there is no visible window between the death of L^2 and the takeover by the tilt.